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# Paleomagnetic correlation of Newark Supergroup volcanics

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## ABSTRACT

**Three Lower Jurassic volcanic sequences are present in the Deerfield, Hartford, and Newark basins of eastern North America. Volcanics from the Deerfield and Newark basins display similar magnetic directions that also agree with magnetizations from associated Lower Jurassic intrusives. The intermediate volcanic sequence, including the Deerfield basalt, the Holyoke basalt, and the base of the Second Watchung basalt, yields anomalous directions that differ by more than 30° from the nominal Early Jurassic direction. These anomalous directions record a short-lived geomagnetic secular variation event that provides a chronostratigraphic marker. The simplest interpretation of the paleomagnetic and geochemical data is a one-to-one correspondence between successive volcanic units of the Hartford and Newark basins, a correlation that agrees with a previous scheme based upon geochemical inferences. Our data suggest that initiation of the second eruptive phase was synchronous over the 300 km from the Deerfield Basin to the Newark Basin. Reevaluation of the widely quoted N1 Early Jurassic paleomagnetic pole for North America indicates that the 72 virtual geomagnetic poles from which it was originally derived are not all independent observations. The N1 reference pole probably represents 7 independent samples of the geomagnetic field, each of which averages at best a small amount of geomagnetic secular variation. As a consequence, the N1 paleomagnetic pole may not represent the time-averaged paleofield.**

## INTRODUCTION

The Newark Supergroup consists of thick continental sedimentary rocks and minor igneous rocks exposed in eastern North America from Nova Scotia to South Carolina (Olsen et al., 1982). The Newark and Hartford basins each contain three distinct volcanic sequences. In the Newark Basin, the First Watchung (Orange Mountain) basalt, the Second Watchung (Preakness) basalt, and the Third Watchung (Hook Mountain) basalt form the volcanic sequence, from bottom to top. The three units in the Hartford Basin are the Talcott, Holyoke, and Hampden basalts, from bottom to top. Each of these "basalts" is actually a sequence of multiple flows. The Deerfield basalt is a single extrusive unit found in the Deerfield Basin to the north of the Hartford Basin.

Paleontologic data suggest that volcanism in the Newark Supergroup and its equivalents is limited to the Hettangian and Sinemurian stages of the Early Jurassic (Olsen et al., 1982), between 208 and 198 Ma (Kent and Gradstein, 1985). K-Ar and $^{40}Ar/^{39}Ar$ ages for volcanic rocks in the Deerfield, Hartford, and Newark basins conflict with the paleontological determinations, and range from 194 to 185 Ma (de Boer, 1968; Armstrong and Besancon, 1970; Sutter and Smith, 1979; Seidemann et al., 1984). Within each basin, even the most reliable ages do not always agree with their relative stratigraphic position, and there is no clear evidence of systematic differences in age between the Newark, Hartford, and Deerfield basins. Isotopic dating of the basalts has been complicated by hydrothermal alteration (Seidemann et al., 1984; Seidemann, 1988; Sutter, 1988). Correlation schemes for volcanics in the Deerfield, Hartford, and Newark basins have been based upon paleomagnetism (de Boer, 1968), geochemistry (Puffer et al, 1981), vertebrate paleontology (Olsen et al., 1982), and isotopic ages (Seidemann et al., 1984). None of these correlations agree completely, and there is no general agreement regarding the relative timing of eruptions from one basin to another. In the course of a paleointensity study of volcanic rocks from the Deerfield, Hartford, and Newark basins, we obtained new paleomagnetic directions that permit a revised paleomagnetic correlation.

## FIELD AND LABORATORY PROCEDURES

We collected 150 oriented cores at 15 localities in the Hartford, Newark, and Deerfield basins, usually from fine-grained basalt at the bottom or top of each flow to ensure magnetic stability for the paleointensity study (Derder et al., 1989). With few exceptions, the tectonic attitude of the flow was measured at the sampling site by using overlying or underlying sedimentary strata. Where direct tectonic attitude measurements were not possible, we used either detailed geologic maps (sites H1B and H3C), previously published tectonic data (site H2A; Balk, 1957), or data provided to us by George Faust of the U.S. Geological Survey (sites from the Second and Third Watchung basalts; 1985, personal commun.).

Laboratory experiments consisted of viscosity measurements, followed by alternating field (AF) demagnetization in fields up to 80 mT and principal component analysis. Thermal demagnetization of representative samples yielded results that were directionally indistinguishable from the AF results; excluding recent viscous magnetizations, the samples yielded simple, single-vector demagnetization trajectories after partial AF demagnetization in peak fields of 5–10 mT. Saturation magnetization measurements ($J_S$-T) were made on 76 samples.

## PALEOMAGNETIC DATA

The Talcott, Holyoke, and Hampden basalts exhibit low magnetic viscosity, typically less than 5%. The site mean directions fall into two groups (Table 1 and Fig. 1a). Sites from the Talcott and Hampden basalts yield identical directions within experimental uncertainty. In contrast, sites collected from the intervening Holyoke basalt yield significantly different directions characterized by a much shallower inclination and more easterly declination, displaced approximately 30° from the Talcott and Hampden basalts. Our new results differ from those presented by de Boer (1968), but they agree with the pioneering work of Irving and Banks (1961).

Results from the Newark Basin volcanics are not as coherent as those of the Hartford Basin volcanics, but we were still able to obtain directions of magnetization that we think are reliable. The difficulties are illustrated by sites W2A1 and W2A2, which come from the same extrusive unit but yield significantly different mean directions (Fig. 1b). Most Newark Basin samples exhibit larger magnetic viscosity than the Hartford Basin basalts; samples from site W2A1 from the base of the flow have an exceptionally large viscosity that varies from 33% to 62%. The best estimate of the paleofield direction is provided by site W2A2 at the top of the Second Watchung basalt, where magnetic viscosity is approximately 6%.

Site mean directions from the Newark Basin volcanics display the same directional variation as the Hartford Basin volcanics. The First Watchung basalt and the Third Watchung basalt have approximately the same direction, but the shallowly inclined direction of the Second Watchung basalt differs by about 30° (cf. Fig. 1, a and b). Our findings are in accord with the results of McIntosh et al. (1985), who found that mean directions from the three Watchung ba-

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salts were approximately the same, with the exception of the base of the Second Watchung basalt where directions close to those of our site W2A2 were observed.

## PALEOMAGNETIC CORRELATION

The most straightforward interpretation of the directional data is a one-to-one correlation of the three extrusive sequences in the Hartford Basin and the Newark Basin (Fig. 2). The lower magnetostratigraphic unit corresponds to the First Watchung basalt and has the same direction of magnetization as the Talcott basalt of the Hartford Basin. The intermediate magnetostratigraphic unit, which includes only the base (first flow?) of the Second Watchung basalt, has an average direction comparable to that of the Holyoke basalt. The upper unit, which includes the upper part (second and younger flows) of the Second Watchung basalt and the Third Watchung basalt, displays an average direction which agrees with that of the Hampden basalt. This correlation is in agreement with a previous correlation based on geochemical data (Puffer et al., 1981).

The Deerfield basalt yields a shallowly inclined direction typical of sites from the intermediate extrusives in the Newark and Hartford basins (Table 1 and Fig. 1a), in agreement with previously published results. Two sites studied by Bowker (1960) display significantly different directions, similar to the results noted above for the sites from the Holyoke basalt and the base of the Second Watchung basalt. For these reasons, we correlate the Deerfield basalt with the intermediate magnetostratigraphic unit in the Newark and Hartford basins, suggesting synchroneity of eruption over a distance of 300 km.

## DISCUSSION

The paleomagnetic results from the extrusives may be compared with data from the Lower Jurassic intrusives in the Hartford and Newark basins because their $^{40}Ar/^{39}Ar$ ages are comparable (Palisades sill, 193 ±9 Ma and 202 ±1 Ma;

TABLE 1. NEW PALEOMAGNETIC DATA

| Site | Flow unit | lat (°N) | long (°W) | n | Inc (°) | Dec (°) | k | $\alpha_{95}$ (°) | VGP lat (°N) | VGP long (°E) |
|---|---|---|---|---|---|---|---|---|---|---|
| | Deerfield basin | | | | | | | | | |
| D | Deerfield basalt | 42.614 | 72.551 | 10 | -6.0 | 10.8 | 258 | 2.8 | 43.4 | 92.6 |
| | Hartford basin | | | | | | | | | |
| H1A | Talcott basalt, first flow | 41.906 | 72.762 | 5 | 30.6 | 2.8 | 325 | 3.5 | 64.4 | 100.9 |
| H1B | Talcott basalt, second flow | 41.724 | 72.812 | 8 | 33.6 | 358.3 | 275 | 3.0 | 66.6 | 111.3 |
| H2A | Holyoke basalt | 42.303 | 72.526 | 9 | -3.3 | 18.5 | 269 | 2.8 | 43.0 | 81.8 |
| H2B | Holyoke basalt | 42.184 | 72.674 | 5 | -6.8 | 12.3 | 739 | 2.3 | 43.1 | 90.4 |
| H2C | Holyoke basalt | 41.902 | 72.761 | 5 | 5.4 | 29.0 | 700 | 2.4 | 43.0 | 65.8 |
| H3A | Hampden basalt, first flow | 42.106 | 72.681 | 8 | 29.6 | 6.0 | 165 | 3.9 | 63.2 | 94.4 |
| H3B | Hampden basalt, second flow | 42.106 | 72.681 | 9 | 32.0 | 1.5 | 191 | 3.6 | 65.2 | 104.0 |
| H3C | Hampden basalt | 41.652 | 72.768 | 10 | 30.4 | 4.8 | 741 | 1.6 | 64.4 | 96.6 |
| | Newark basin | | | | | | | | | |
| W1A | First Watchung basalt | 40.667 | 74.397 | 4 | 23.8 | 4.0 | 102 | 6.9 | 61.6 | 97.4 |
| W1B | First Watchung basalt | 40.633 | 74.447 | 9 | 26.9 | 354.0 | 222 | 3.1 | 63.1 | 118.6 |
| W2A1 | Second Watchung basalt, first flow | 40.886 | 74.224 | 5 | -3.1 | 3.1 | 136 | 5.4 | 47.5 | 101.1 |
| W2A2 | Second Watchung basalt, first flow | 40.893 | 74.220 | 5 | -4.4 | 14 | 266 | 3.8 | 45.1 | 85.8 |
| W3A | Third Watchung basalt, first flow | 40.679 | 74.539 | 5 | 35.9 | 6.1 | 288 | 4.8 | 68.6 | 89.7 |
| W3B | Third Watchung basalt,second flow | 40.679 | 74.539 | 10 | 34.4 | 359.4 | 366 | 2.4 | 68.2 | 106.9 |

Note: n: number of samples collected at each site; Inc, Dec: tectonically corrected site mean inclination and declination; k: precision parameter; $\alpha_{95}$: radius of confidence circle (P = 0.05); VGP lat, long: latitude and longitude of virtual geomagnetic pole.

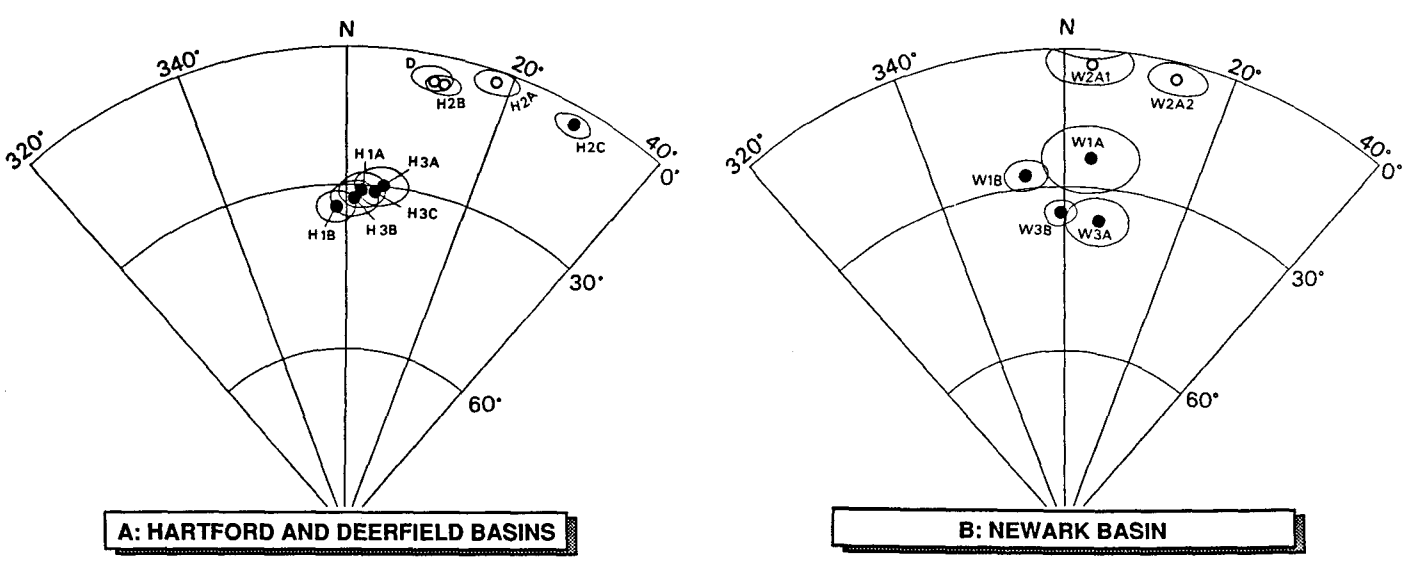



**Figure 1. Equal-area projections of new site mean directions for Newark Supergroup volcanics, with associated confidence circles (P = 0.05). Site labels as in Table 1; *solid symbols represent positive inclination, open symbols represent negative inclination.***

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Mt. Carmel sill, 191 ±4 Ma; Hartford Basin flows, 187 ±3 Ma; ages from Dallmeyer, 1975; Sutter and Smith, 1979; Seidemann, 1988; Sutter, 1988). Mean virtual geomagnetic poles (VGP) from the lower and upper extrusives are indistinguishable from the intrusives. In contrast, the intermediate extrusive yields VGP that are not found in any of the intrusives. This suggests that the intermediate unit records a short-lived departure of the field from its nominal direction, a departure whose rapidity is attested to by higher between-site dispersion.

The observed directional fluctuation may be due either to secular variation or to a geomagnetic excursion. Derder et al. (1989) favored the first interpretation because (1) paleointensity estimates from the Holyoke basalt and Palisades differ by only 13%, well within the range of more recent geomagnetic secular variation; (2) excursions are usually characterized by a large field intensity decrease (Coe et al., 1984); and (3) an explanation in terms of transitional directions recorded during a polarity transition is implausible because reversals have not been reported from the interbedded sedimentary strata (McIntosh et al., 1985).

Whether the directional fluctuation is due to secular variation or to a geomagnetic excursion, it constitutes a useful magnetostratigraphic marker, suggesting simultaneity of the eruptions that produced the Deerfield and Holyoke basalts in their entirety, and the initial stages of the eruptions that produced the Second Watchung basalt. The eruptions may have persisted over a longer interval in the Newark Basin, as suggested by the observation of nominal Early Jurassic field directions in the upper part of the Second Watchung basalt.

The coincidence of directions between the intermediate extrusive units does not conclusively prove that these units are time equivalents. The patterns of geomagnetic secular variation are not known or predictable, and thus it is possible (but unlikely) that the directional fluctuations observed in these three volcanic sequences represent completely unrelated geomagnetic events.

Contrary to a suggestion made by de Boer (1968), the paleomagnetic data now available do not require different timing for eruptions of the lower or the upper volcanics in the Hartford and Newark basins. The similarity in magnetization direction between the Palisades sill (Rigotti, 1976) and interconnected volcanic flows (Kodama, 1983) suggests that these units may be correlated with either the lower or the upper extrusive sequences in the Hartford and Newark basins, but not the intermediate sequence. Ratcliffe (1988) used geochemical data to show that the Palisades sill is a feeder for either the lower or intermediate extrusive sequence. Taken together, the geochemical and paleomagnetic results suggest a correlation between the Palisades sill and the lower extrusive sequence.

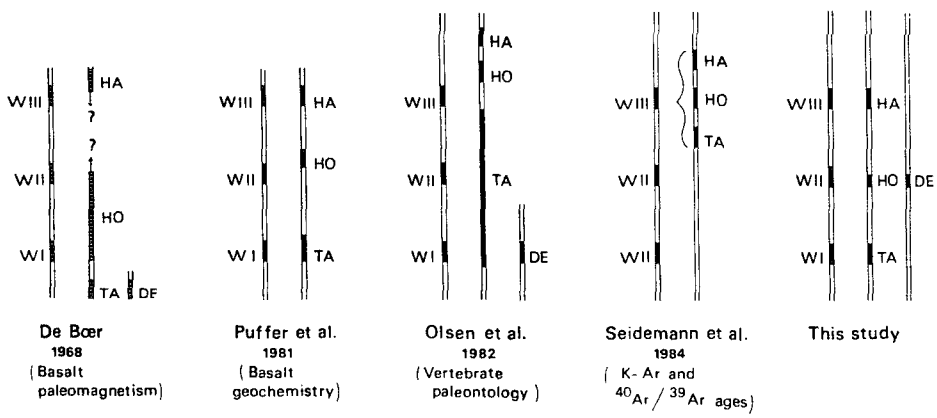



**Figure 2. Proposed correlation schemes for Lower Jurassic volcanics (black regions) in Newark, Hartford, and Deerfield basins. WI, WII, WIII—Watchung basalts; TA—Talcott basalt; HO—Holyoke basalt; HA—Hampden basalt; DE—Deerfield basalt.**

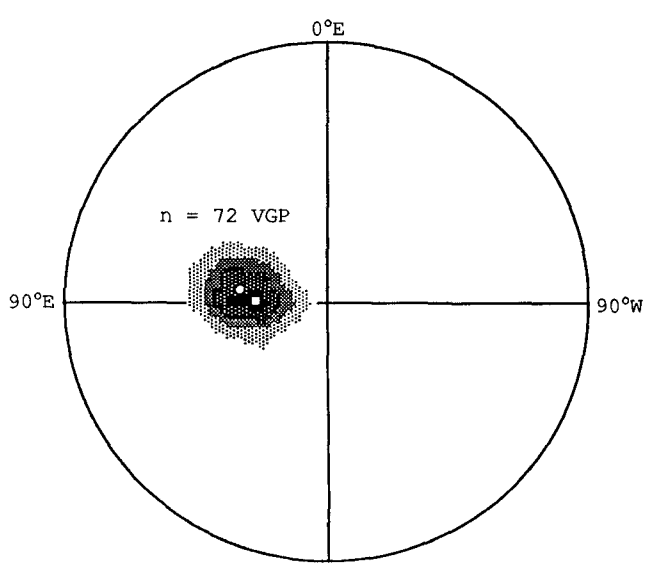



**Figure 3. Equal-area contour plot of 72 virtual geomagnetic poles (VGP) used in calculating N1 pole of Smith and Noltimier (1979). North polar projection; contour interval 5σ using Kamb's (1959) technique. White circle—mean VGP calculated by Smith and Noltimier (1979), which includes VGP from intermediate extrusive. White square—mean VGP from Table 2, which excludes VGP from intermediate extrusive. Note that circle lies south and east of mode, but square is good approximation of mode.**

## EARLY JURASSIC REFERENCE POLE FOR NORTH AMERICA

Smith and Noltimier (1979) divided paleomagnetic data from the Newark trend igneous rocks into two groups on the basis of isotopic age and directional grouping. Their N1 pole (lat 63.0°N, long 83.2°E) has been widely quoted in tectonic interpretations of the North American apparent polar wander path, but 16 of the 72 sites (22%) that they used to define this pole come from the anomalous flows described above. Figure 3 illustrates the bias caused by inclusion of these sites.

We have recalculated the N1 pole position by (1) excluding VGP from the Deerfield basalt, the Holyoke basalt, and the lower part of the Second Watchung basalt; (2) including only results from contemporary studies where demagnetization and analytical methods are more complete; and (3) excluding site mean directions whose $\alpha_{95}$ exceeds 10°. Because the three extrusives in the Hartford and Newark basins represent time-equivalent eruptions, their site VGP were averaged to create a mean VGP for each extrusive. Because each "basalt" is actually more than one flow, its mean VGP may average some geomagnetic secular variation, but even the occurrence of multiple flows does not guarantee that a significant time period has been recorded (McWilliams et al., 1982). Accordingly, we have chosen a conservative approach in which each of the three extrusive mean VGP was given unit weight in the overall average.

The extrusive mean VGP were then combined with mean VGP from four intrusive units in the Hartford and Newark basins, yielding seven independent observations that define the N1 pole (Table 2). As described above, the Palisades sill may be contemporaneous with either the lower or upper extrusive sequence. If this is correct, then the number of independent obser-

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TABLE 2. VIRTUAL GEOMAGNETIC POLES FROM THE HARTFORD AND NEWARK BASINS

| Unit | Reference | VGP lat (°N) | VGP long (°E) | N | R | K | $A_{95}$ |
|---|---|---|---|---|---|---|---|
| Extrusive units | | | | | | | |
| Lower | This paper and McIntosh et al. (1985) | 64.8 | 107.7 | 8 | 7.9760 | 291 | 2.9 |
| Middle (lower part)* | This paper and McIntosh et al. (1985) | 47.1 | 82.3 | 10 | 9.8705 | 69 | 5.8 |
| Middle (upper part) | This paper and McIntosh et al. (1985) | 68.4 | 84.7 | 5 | 4.9421 | 69 | 9.3 |
| Upper | This paper and McIntosh et al. (1985) | 64.9 | 97.7 | 9 | 8.9658 | 233 | 3.4 |
| Extrusive mean | | 66.3 | 97.3 | 3 | 2.9923 | 260 | 5.0 |
| Intrusive units | | | | | | | |
| East Rock sill | Smith and Noltimier (1979) | 67.3 | 70.5 | 6 | 5.9855 | 345 | 3.1 |
| West Rock sill | Smith and Noltimier (1979) | 66.7 | 85.0 | 8 | 7.9014 | 71 | 5.9 |
| Mount Carmel sill | Smith and Noltimier (1979) | 64.8 | 79.1 | 3 | 2.9860 | 143 | 6.8 |
| Palisades sill† | Rigotti (1976) | 74.4 | 97.5 | | | | |
| Intrusive mean | | 68.9 | 81.4 | 4 | 3.9858 | 212 | 4.8 |
| Overall Mean | | 68.0 | 88.6 | 7 | 6.9669 | 181 | 3.9 |

Note: VGP lat, long: latitude and longitude of virtual geomagnetic pole; N: number of sites; R: length of resultant vector; K: precision parameter estimate; $A_{95}$: radius of confidence circle (P = 0.05).
*lower part of middle extrusive unit excluded from mean (see text for details).
†unit weight given to specimens in this analysis, N = 275.

vations should be reduced to six. However, because the VGP from the Palisades sill differs by approximately 10° from the upper and lower extrusive VRP, we think this reduction is probably not required.

Some of the intrusives in the Mesozoic basins of eastern North America appear to have been emplaced almost contemporaneously over a wide area (Sutter, 1988). The average K value for the extrusives and intrusives is 174, considerably larger than the expected K of about 20 for 16° paleolatitude (Irving and Pullaiah, 1976). We think it is quite possible that little geomagnetic secular variation has been recorded by each of the extrusive and intrusive units. In the most conservative interpretation, the N1 pole is defined by a maximum of seven independent cooling units, which may be insufficient to average secular variation reliably. It follows that the weight assigned to the N1 pole in tectonic interpretations of the North American apparent polar wander path should be adjusted accordingly.

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